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Windows Lab

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Join Win11Client and IIServer VMs to Your Team

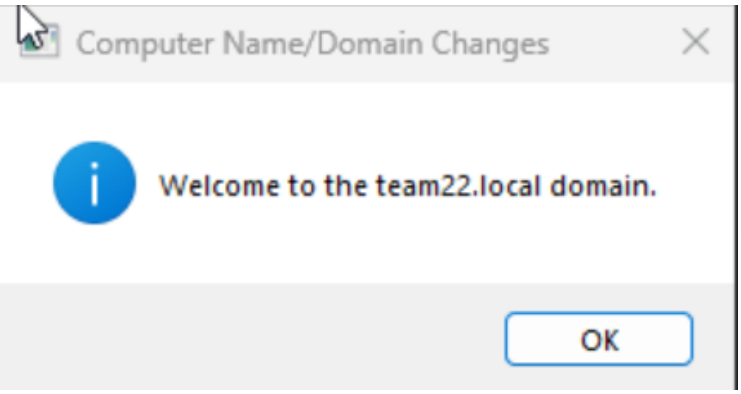

Figure 1: Confirmation message displayed after successfully joining the Win11Client virtual machine to the team22.local Active Domain.

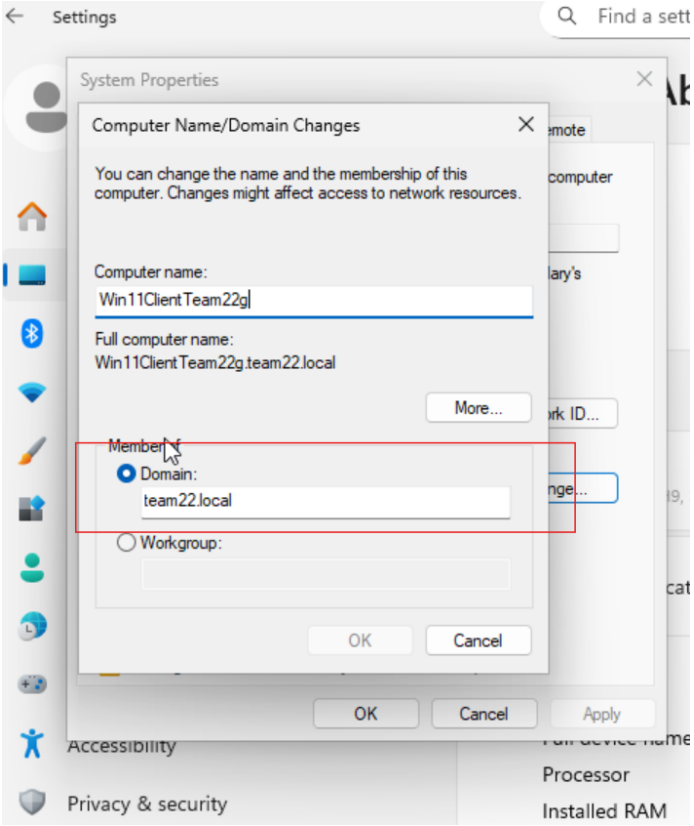

Figure 2: Computer Properties window showing the Win11Client VM successfully configured as a member of the team22.local domain.

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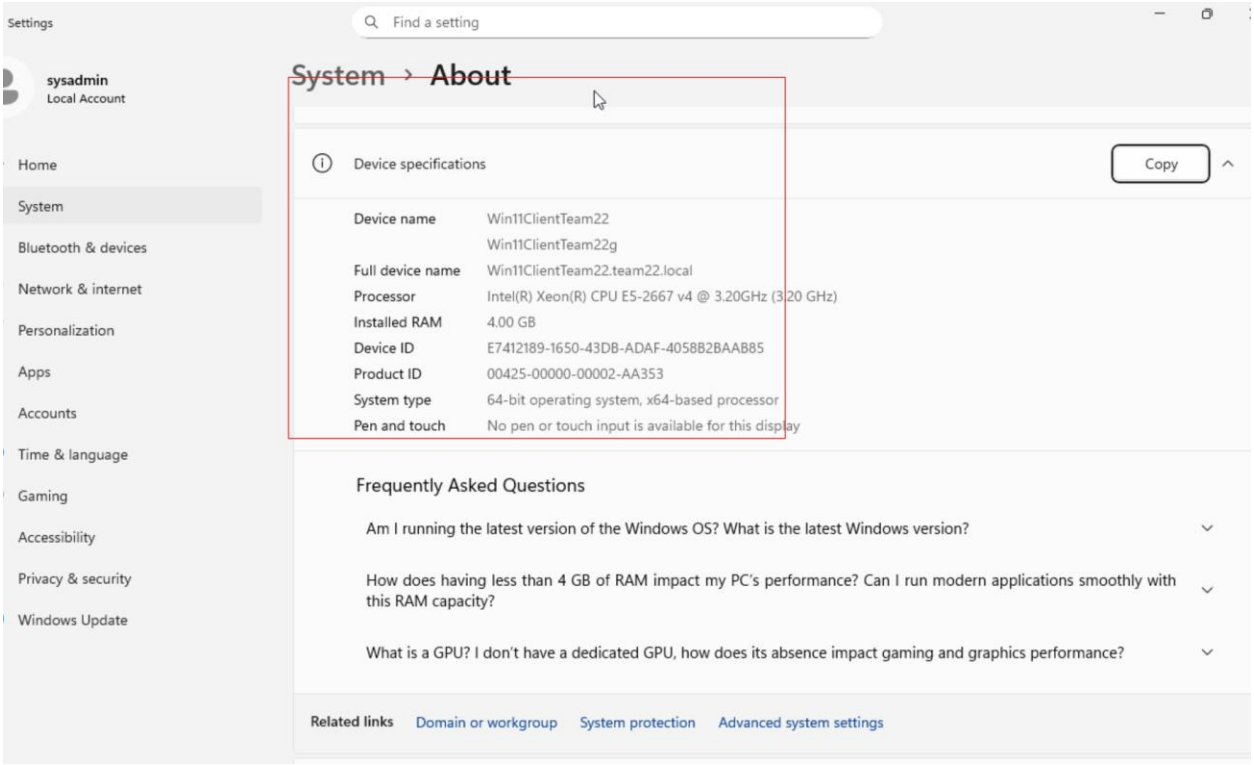

Figure 3: System Information page on the Win11Client VM confirming the computer name and successful domain connection to team22.local

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Create Two Users On Your Team Domain

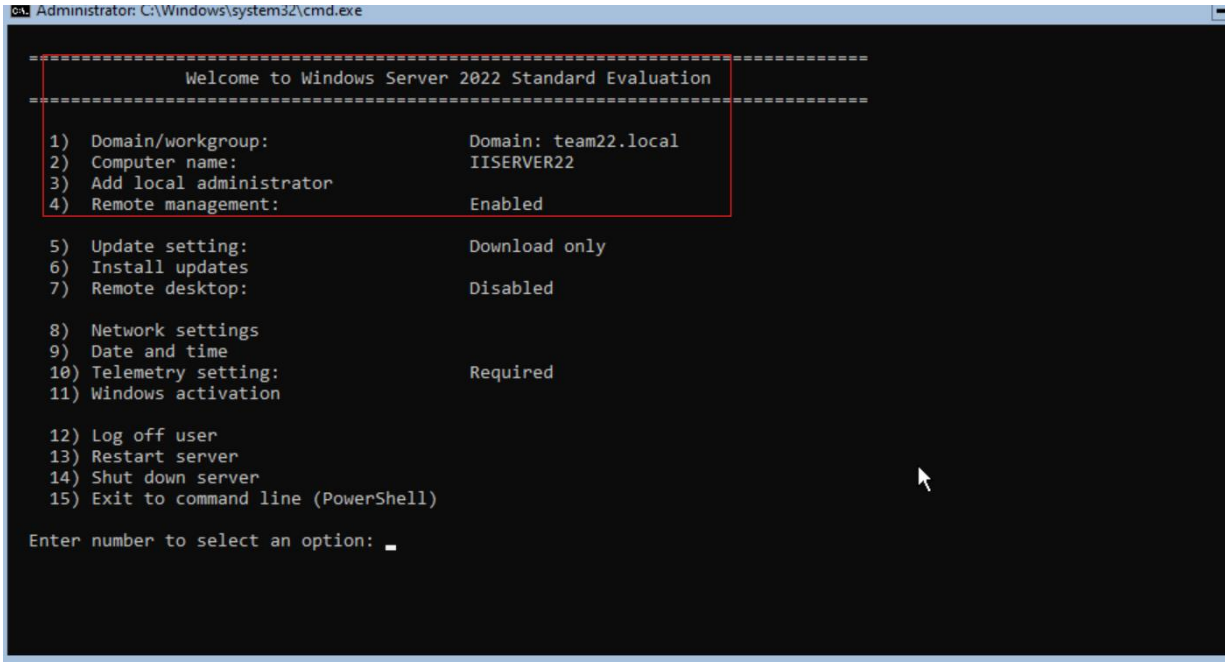

Figure 4: Server configuration screen on IIServer22 confirming that the server is joined to the team22.local domain with remote management enabled.

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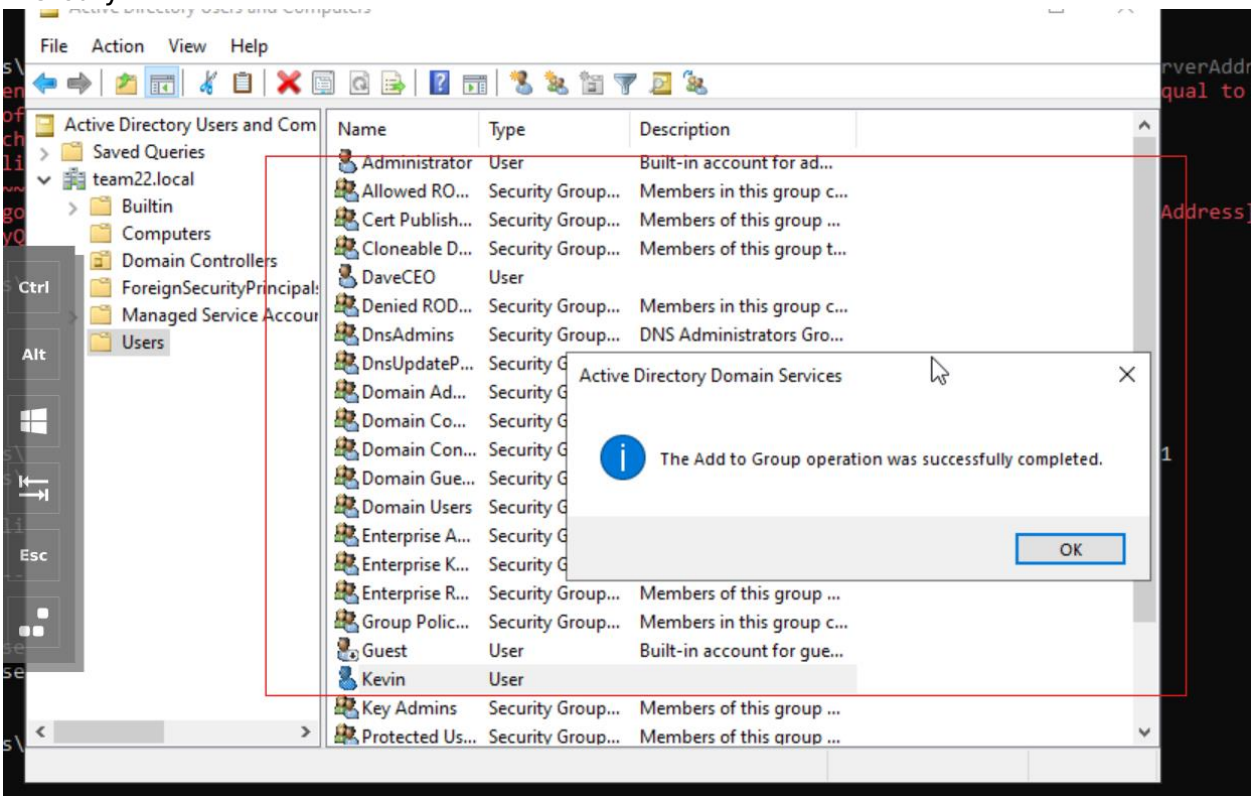

Figure 5 Confirmation message in Active Directory Users and Computers showing successful completion of adding a user to a security group within the team22.local domain.

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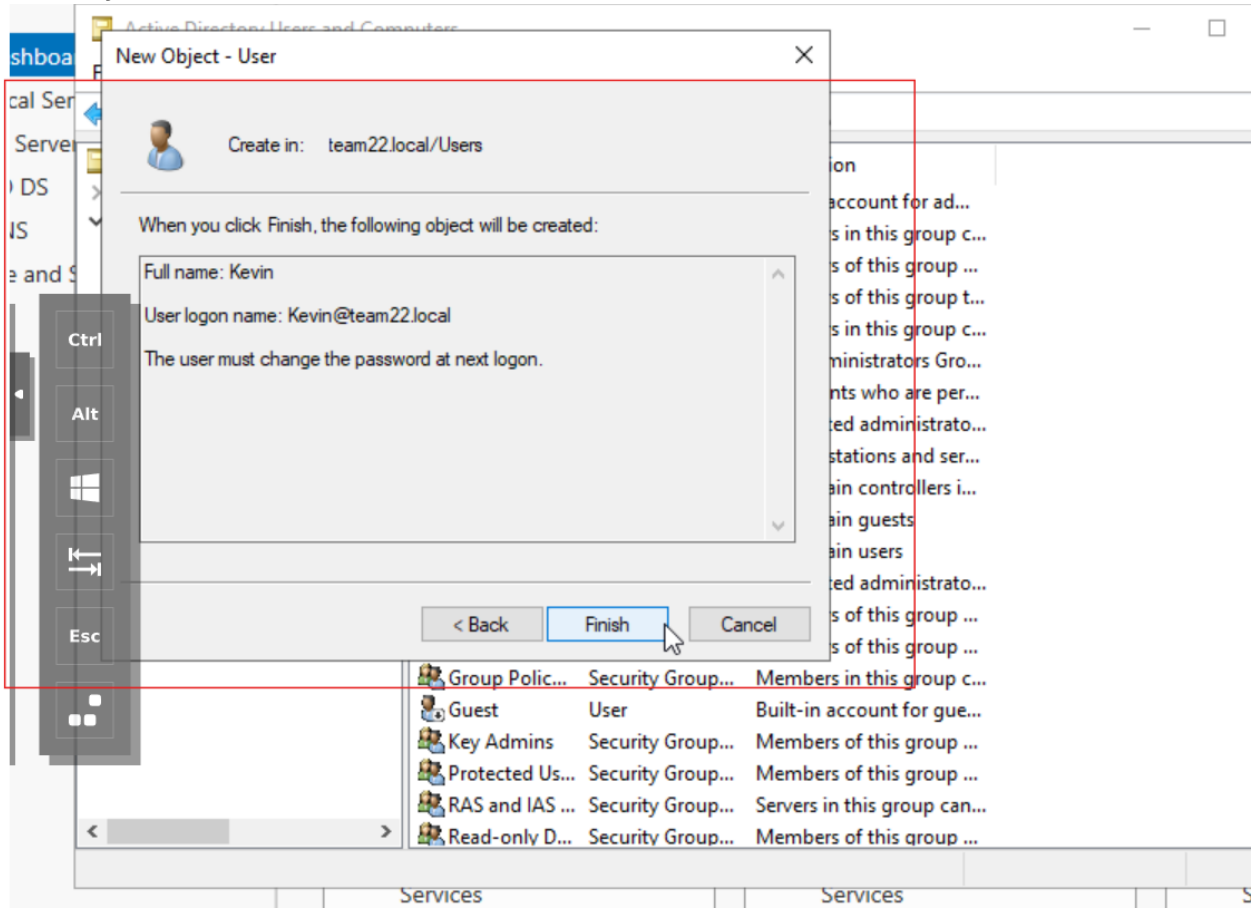

Figure 6: Creating a new user account named Kevin in Active Directory Users and Computers under the team22.localdomain

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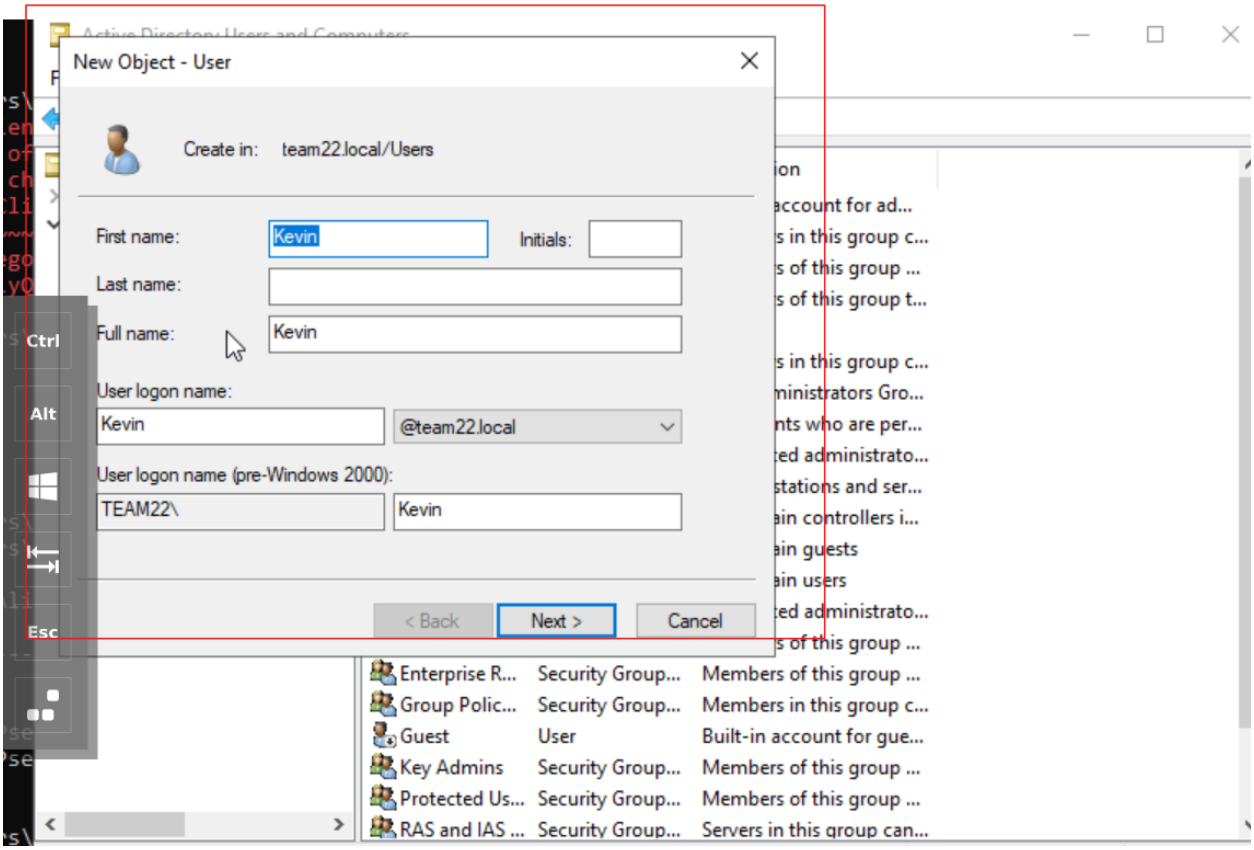

Figure 7: Creating the user account Kevin in the team22.local domain within Active Directory Users and Computers specifying the username and domain login details.

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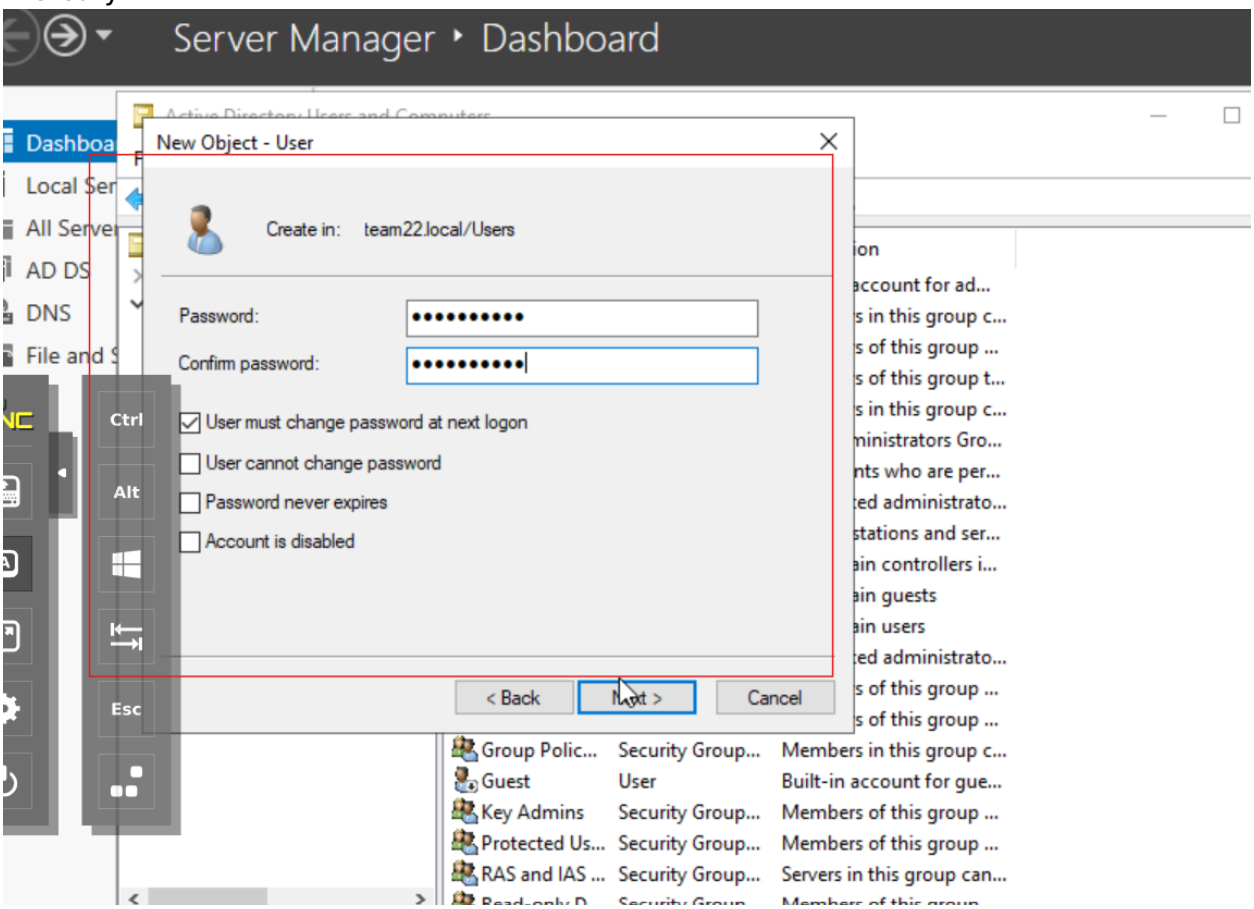

Figure 8: Setting an initial password and configuring login requirements for Kevin user account during creation in Active Directory Users and Computers.

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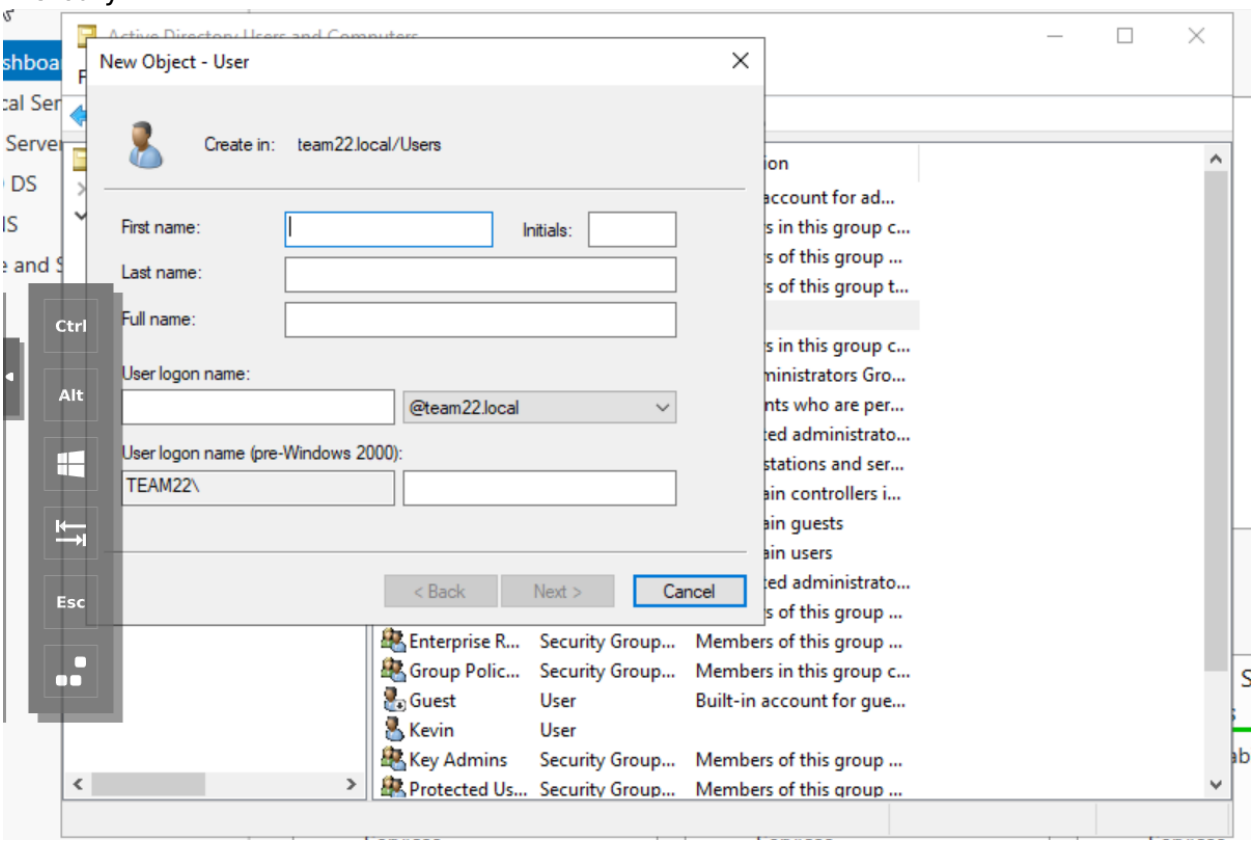

Figure 9: Beginning the process of creating a new Active Directory user by entering basic information and assigning a domain login under team22.local

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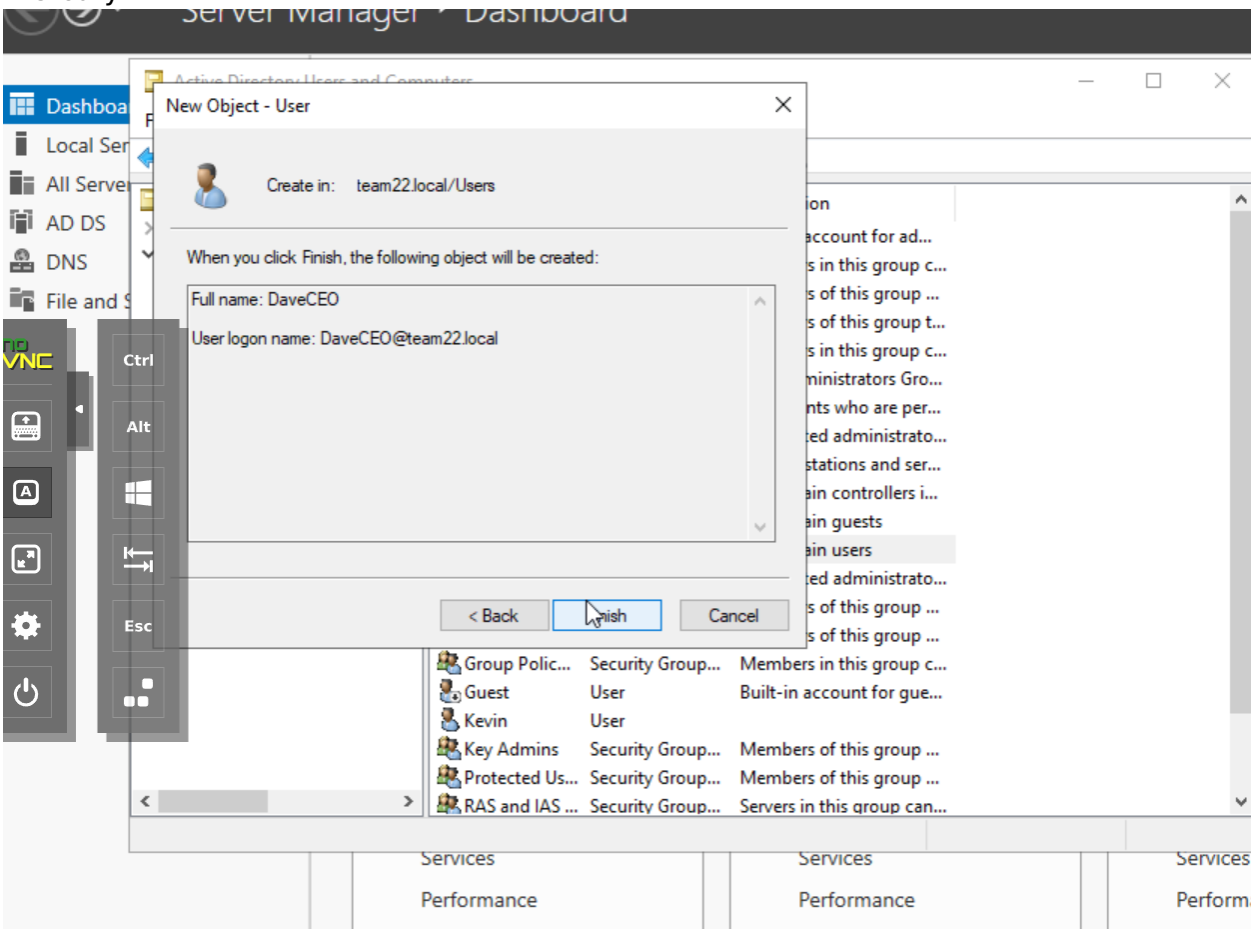

Figure 10: Creating the second user account DaveCEO in the team22.local domain through Active Directory Users and Computers

Add IIServer to the ADServer Server Pool

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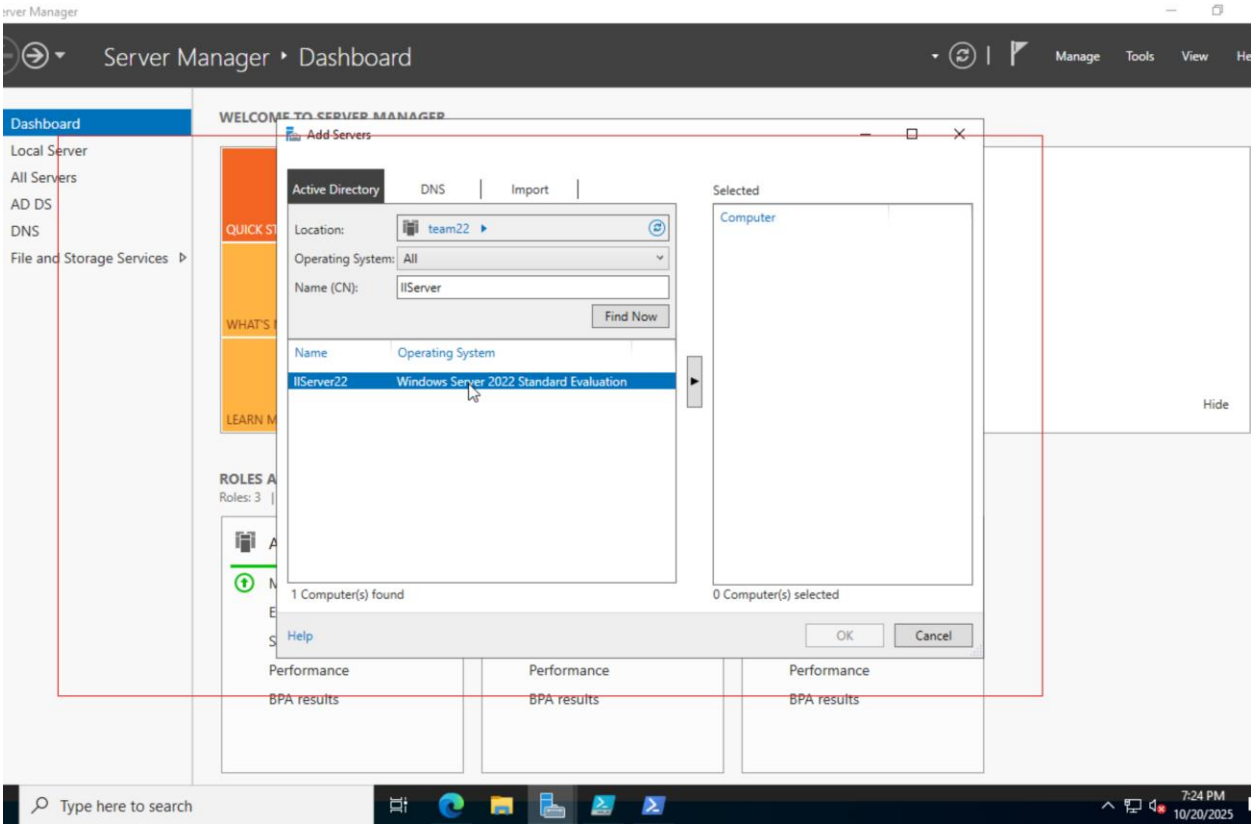

Figure 11: Adding IIServer22 to the Active Directory server pool within Server Manager showing successful detection of the Windows Server 2022 Standard Evaluation System.

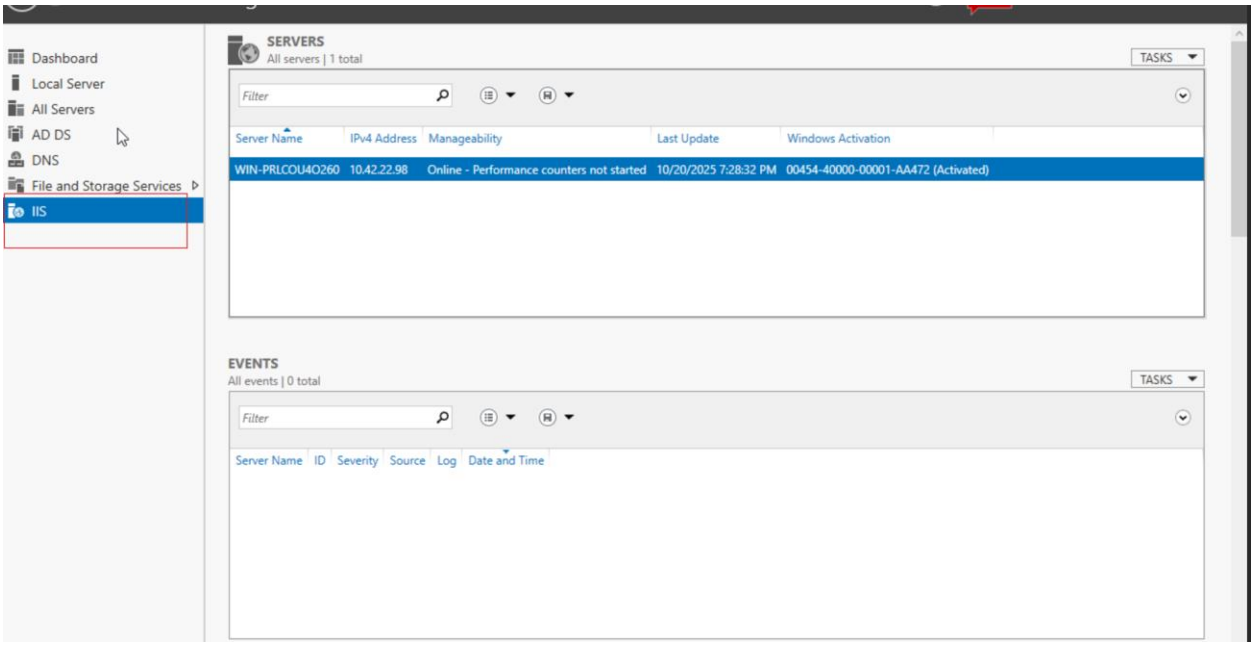

Figure 12: Verifying that IIServer22 has been successfully added and is online within the Server Manager under File and Storage Services IIS, confirming proper connection to the domain network.

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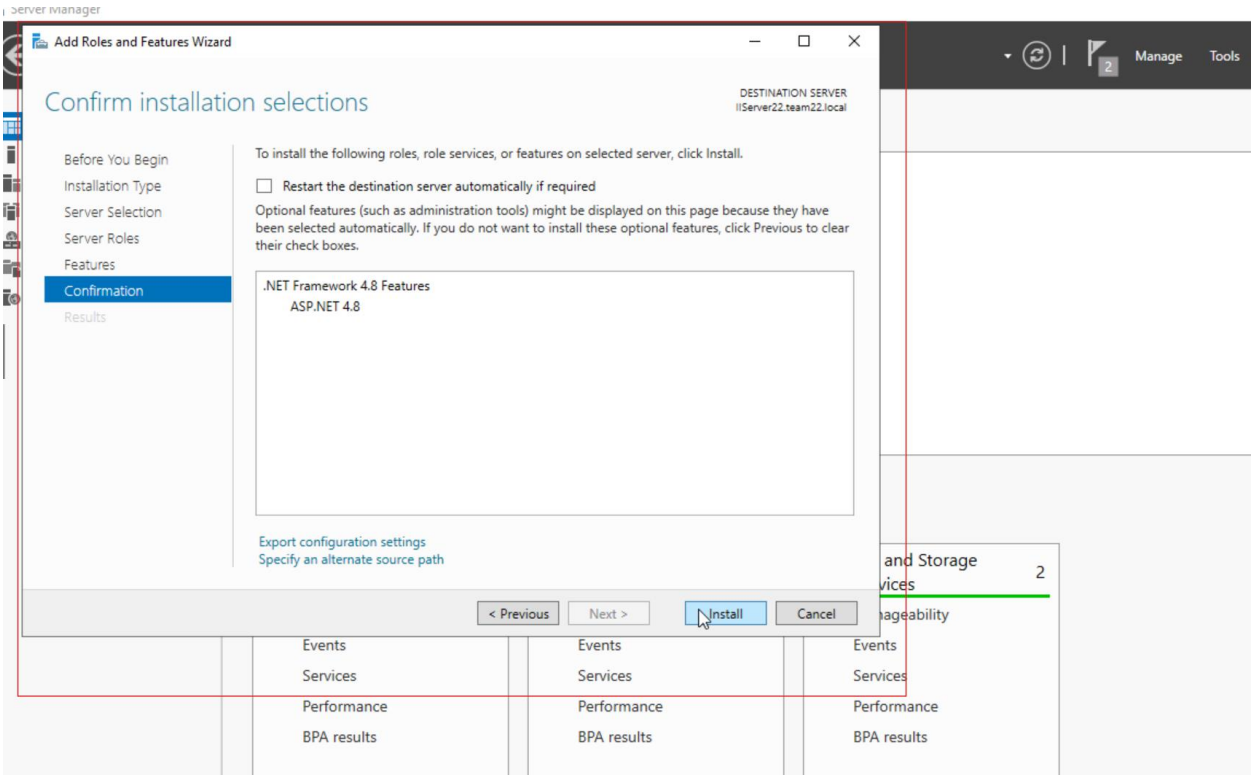
Install an Internet Information Services Web Server on IIServer



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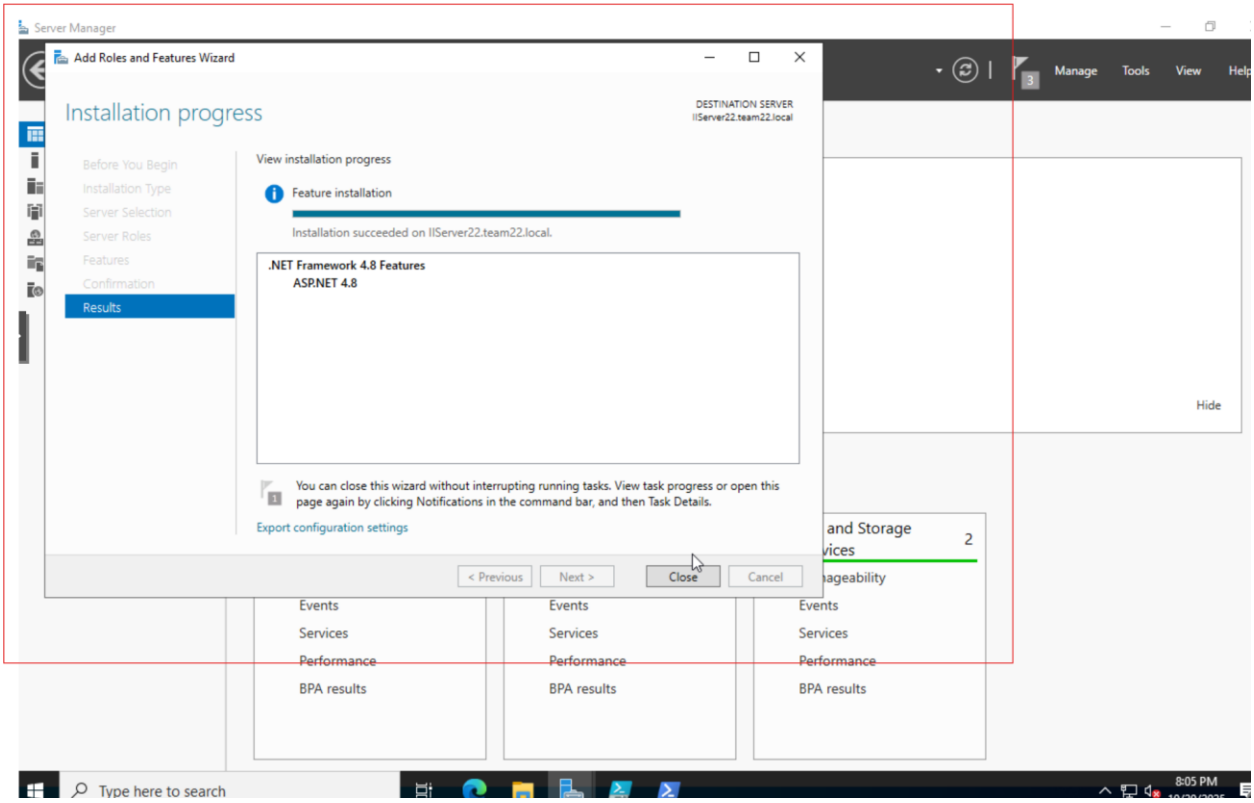

Figure 13: Confirming installation selections for the IIS Server role, including the .NET Framework 4.8 and ASP.NET 4.8 features before proceeding with the installation in Server Manager

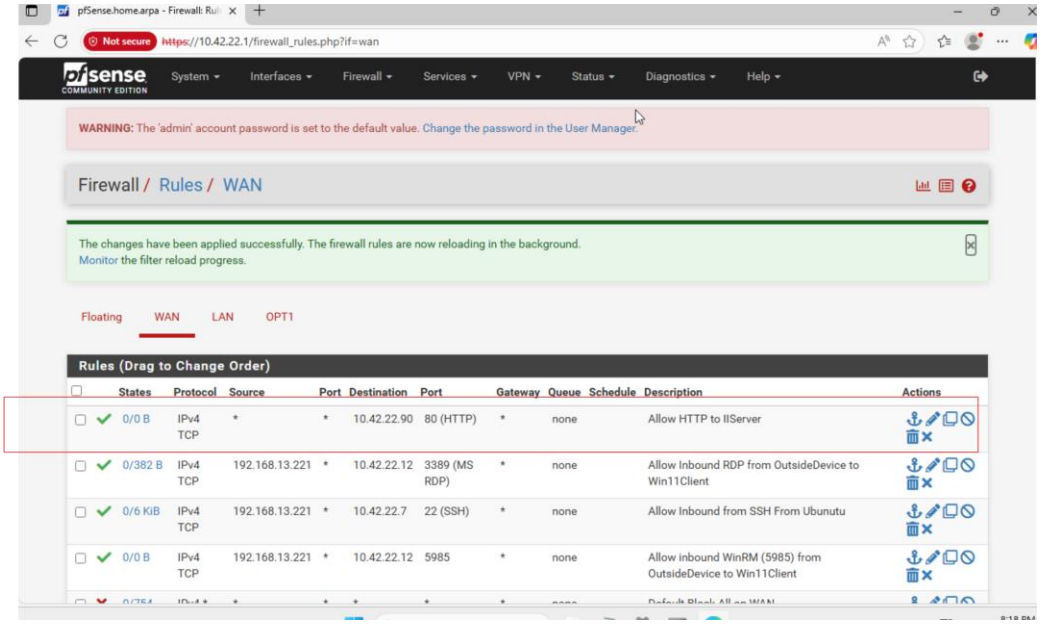

Figure 14: Configured firewall rules in pfSense under the WAN tab, showing allowed inbound traffic for HTTP to IIS Server RDP and WinRM access for Win11Client and SSH access from UbuntuClient.

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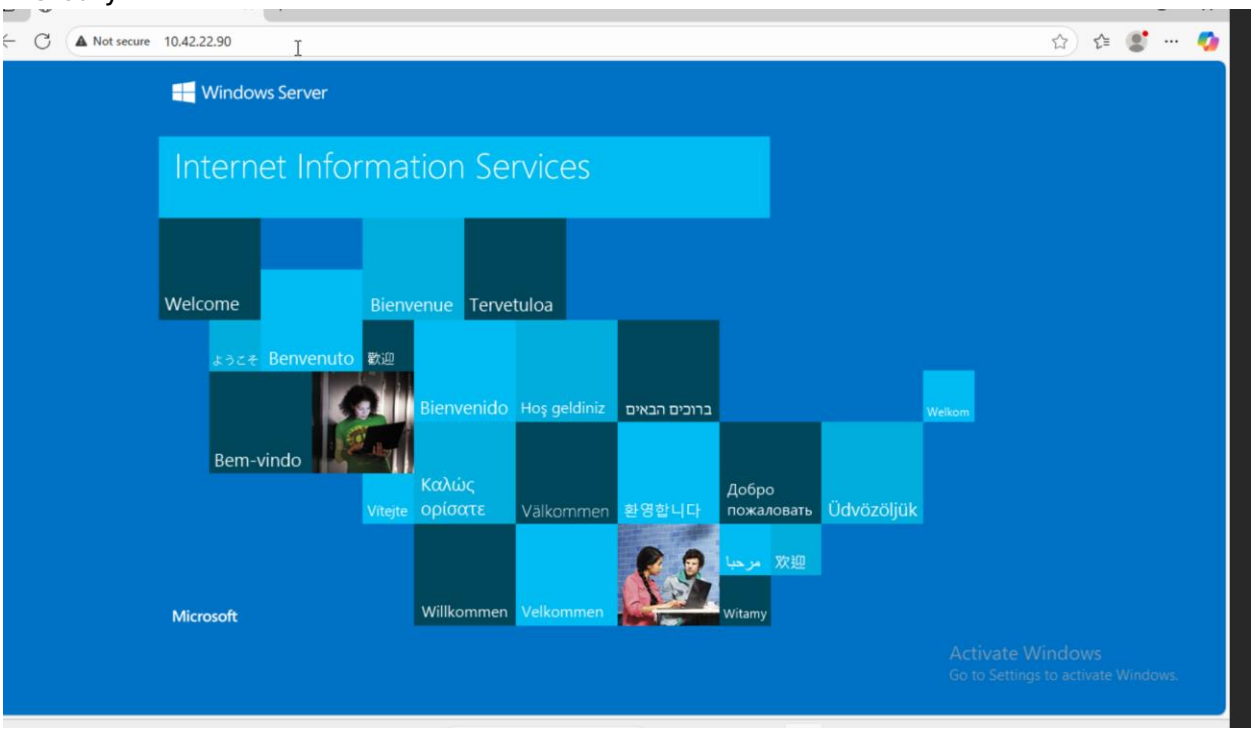

Figure 15: Successful IIS web server deployment verified by accessing the default internet information Services (IIS) webpage through the Server's IP address (10.42.22.90), confirming that IIS is running correctly.

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Create Groups

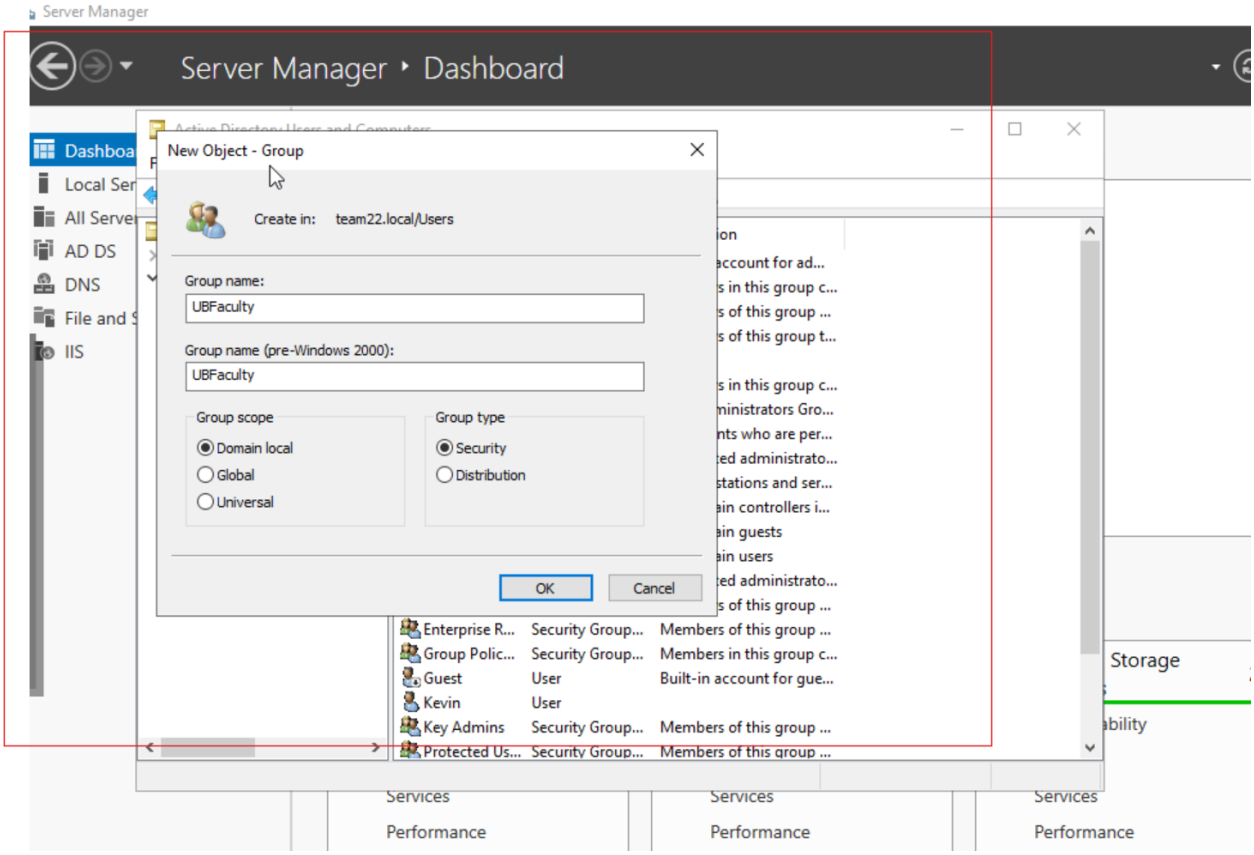

Figure 16: Creating a new Active Directory security group named UBFaculty within the team22.local/Users organizational unit, configured with Global scope and Security type.

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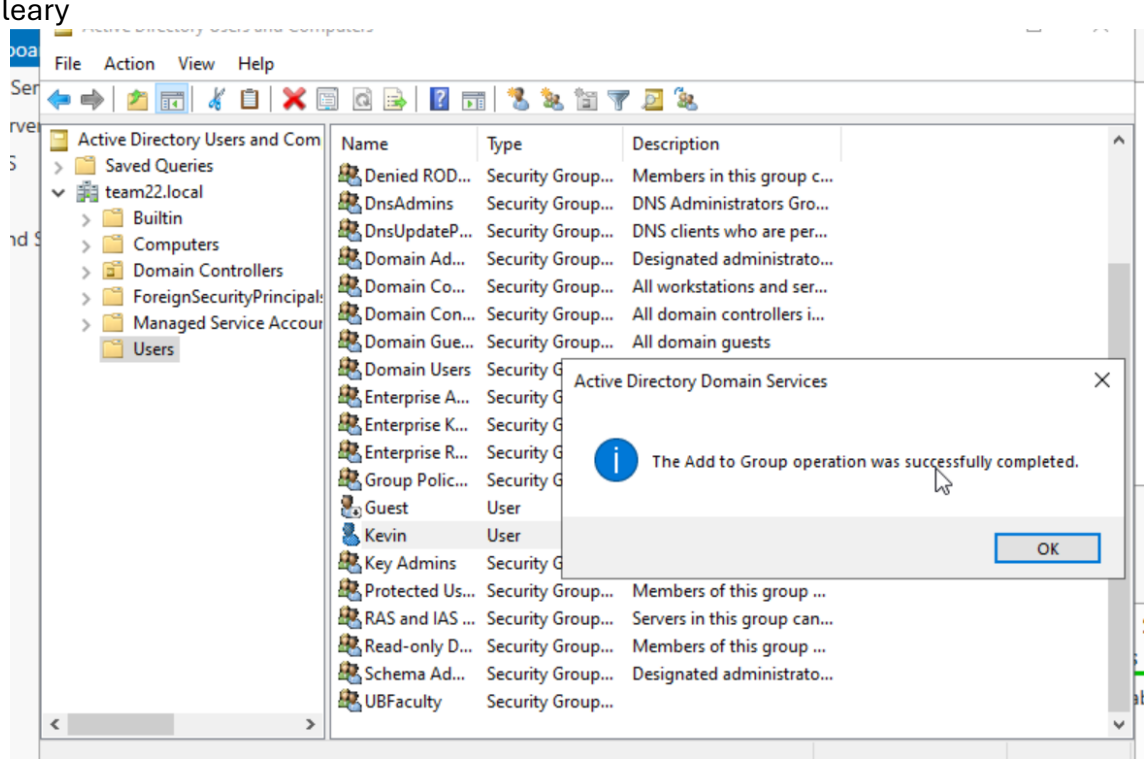

Figure 17: Confirmation message showing that a user was successfully added to the UBFaculty security group within Active Directory under the team22.local domain.

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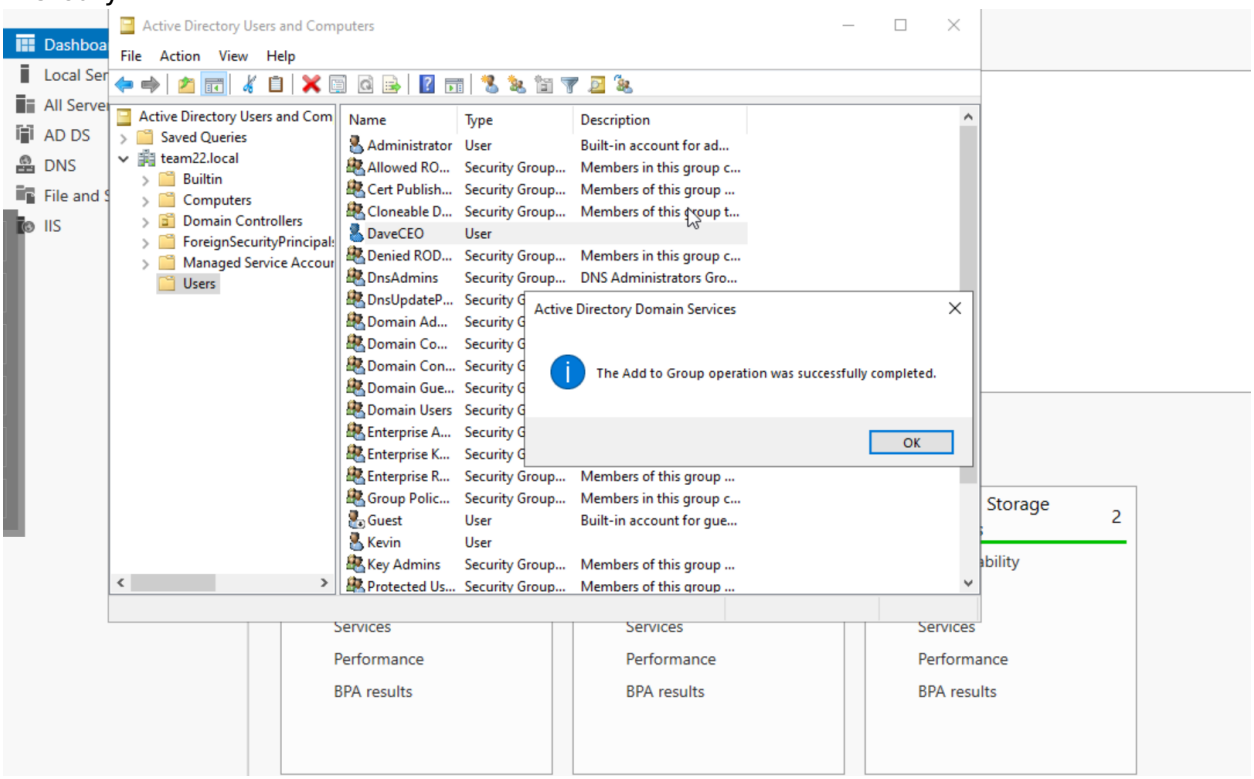

Figure 18: Final verification of group membership in Active Directory showing the membership in Active Directory showing the UBFaculty security group was successfully created and configured under the team22.local domain.

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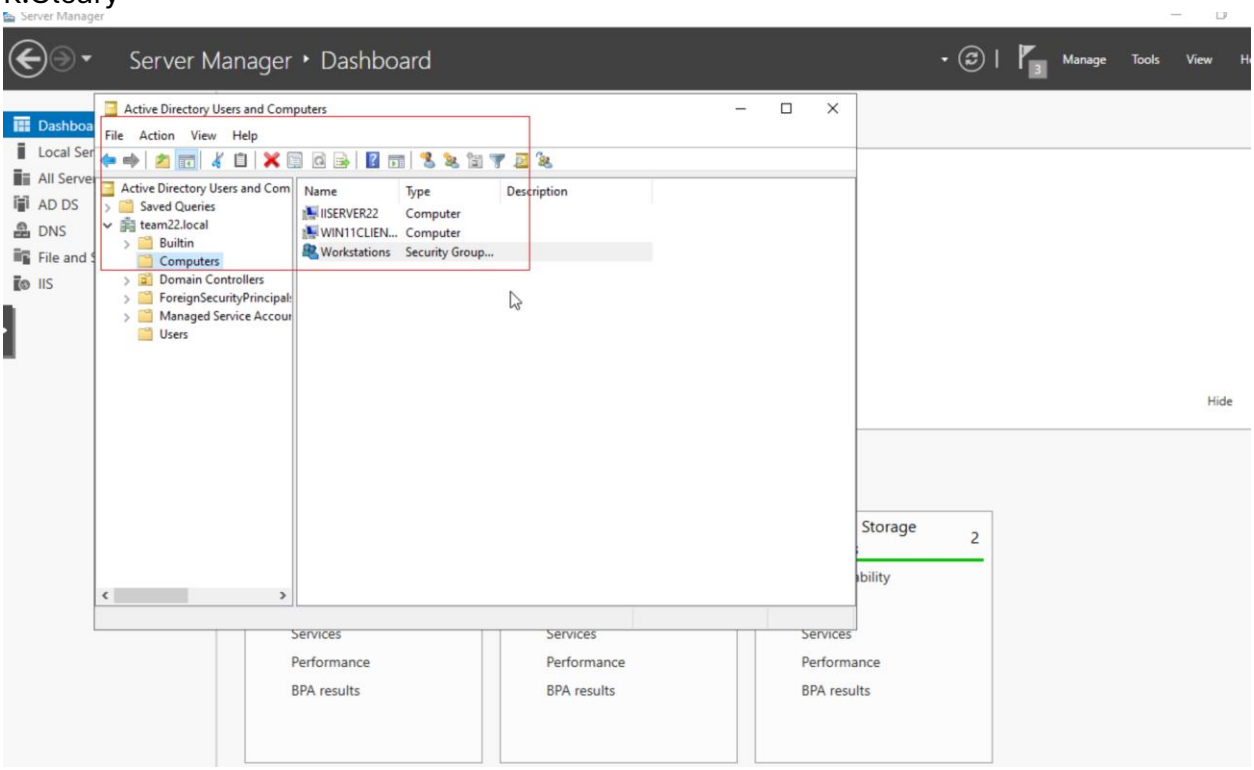

Figure 19: Active Directory view displaying the Computers container, confirming that both domain-joined devices (IIServer22 and Win11ClientTeam22) are successfully registered within the team22.local domain.

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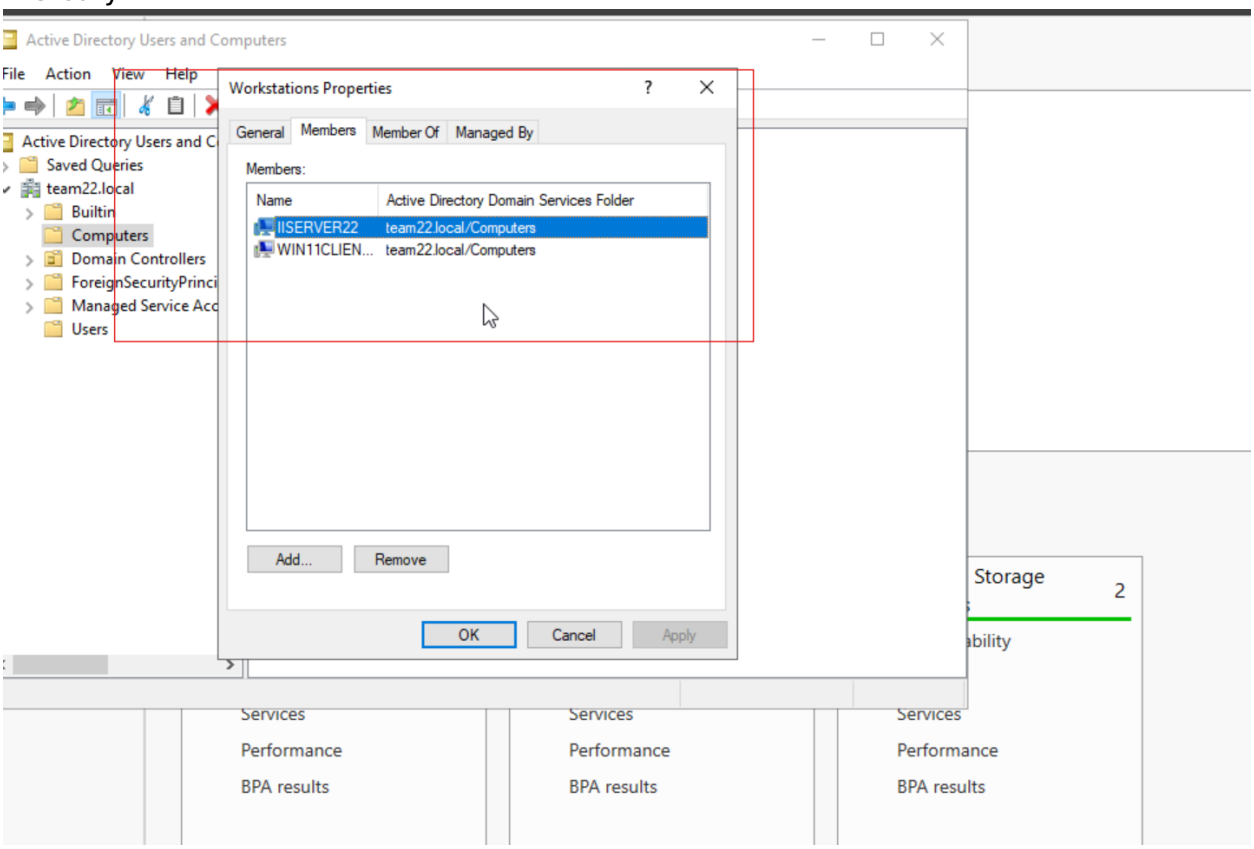

Figure 20: Workstation Properties window showing that both domain-joined devices (IIServer22 and Win11ClientTeam22) are listed as members under Computers container in Active Directory within the team22.local domain.

Enforce a Background Group Policy

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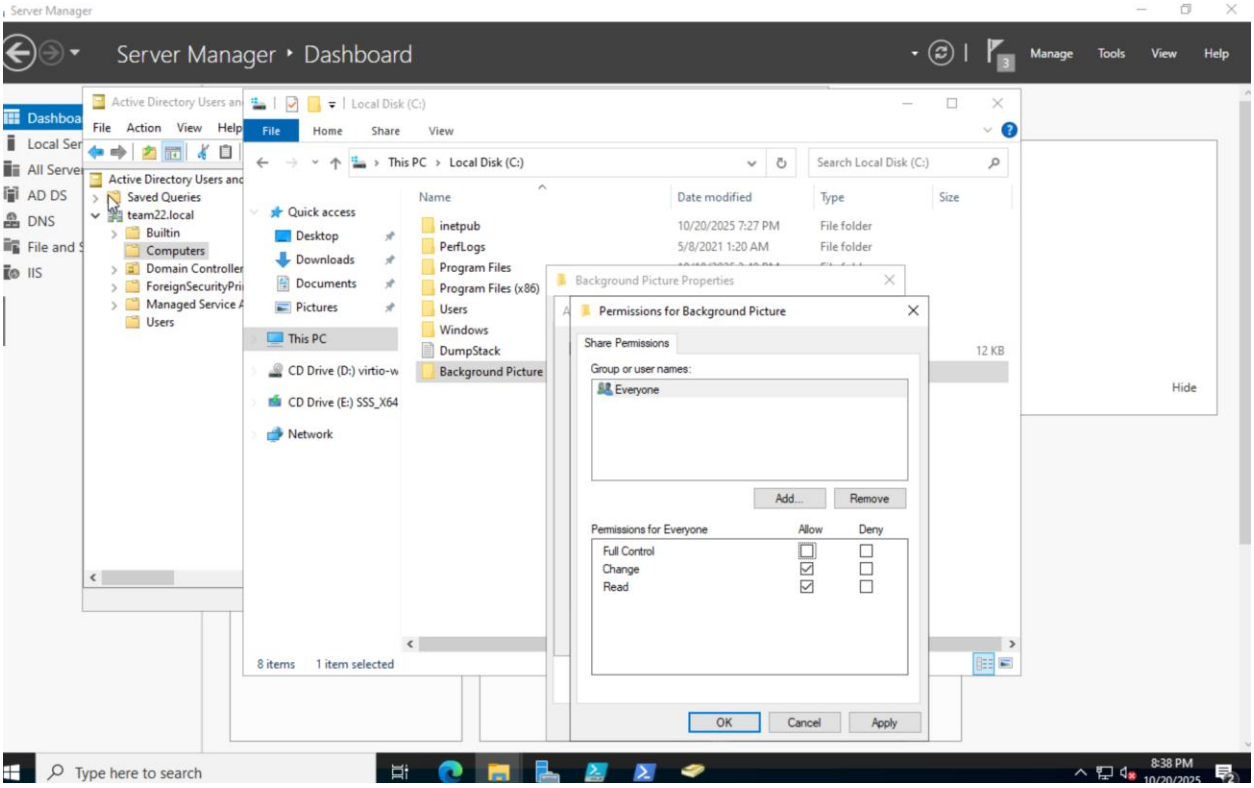

Figure 21: Setting folder permissions for the shared directory Background Picture on the ADserver to allow network access for domain users this folder will be used to store shared resources accessible through Group Policy configurations.

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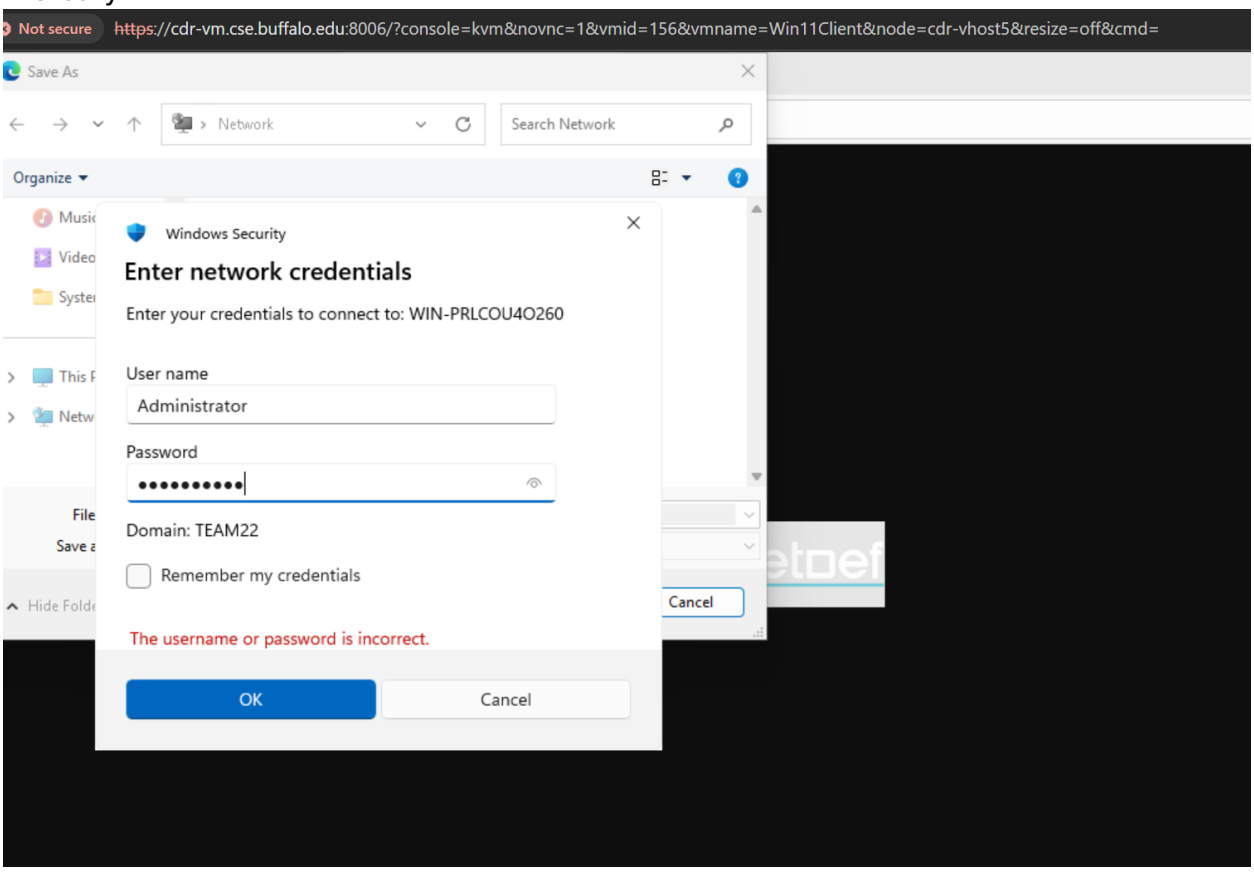

Figure 22: Attempting to access the shared folder from the Win11Client VM using dmain credentials. The login prompt indicates an incorrect username or password showing that authentication failed for the entered domain account (Team22\Administator)

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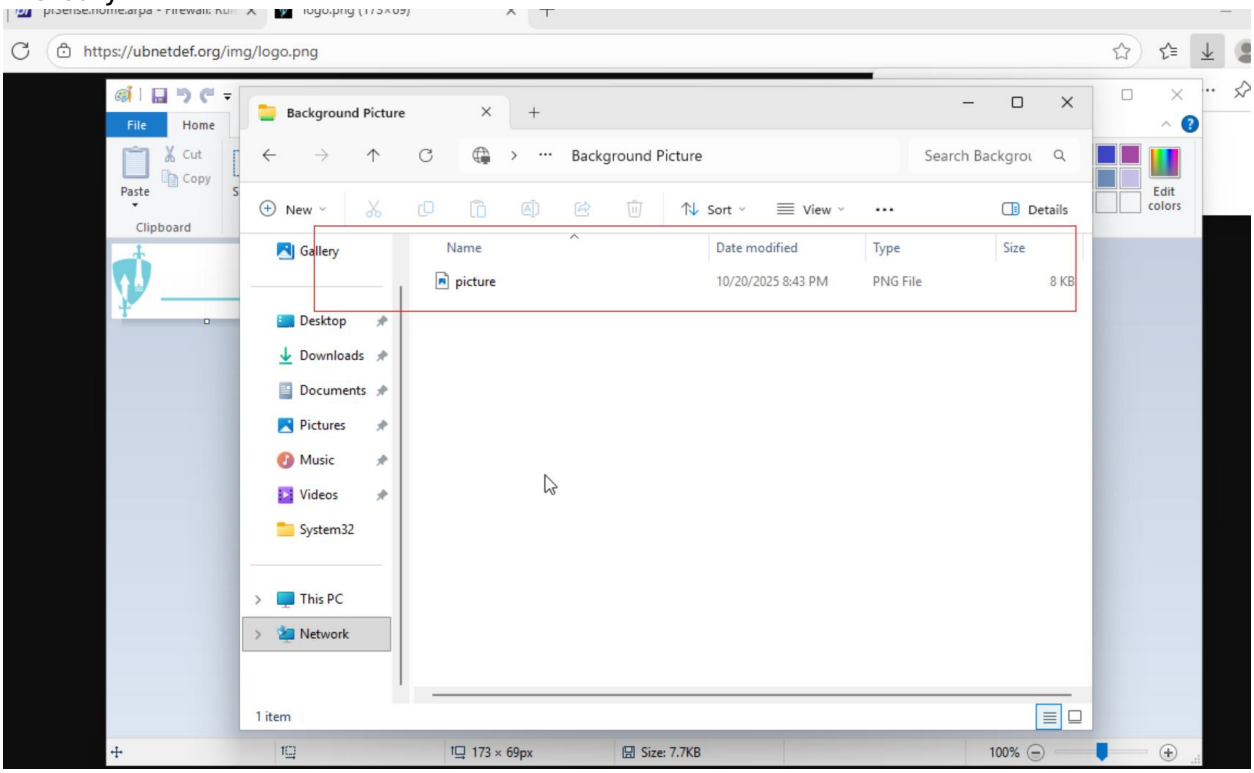

Figure 23:Successful access to the Shared folder Background Picture from the Win11Client VM, confirming proper network share permissions and connectivity between domain devices within team22.local

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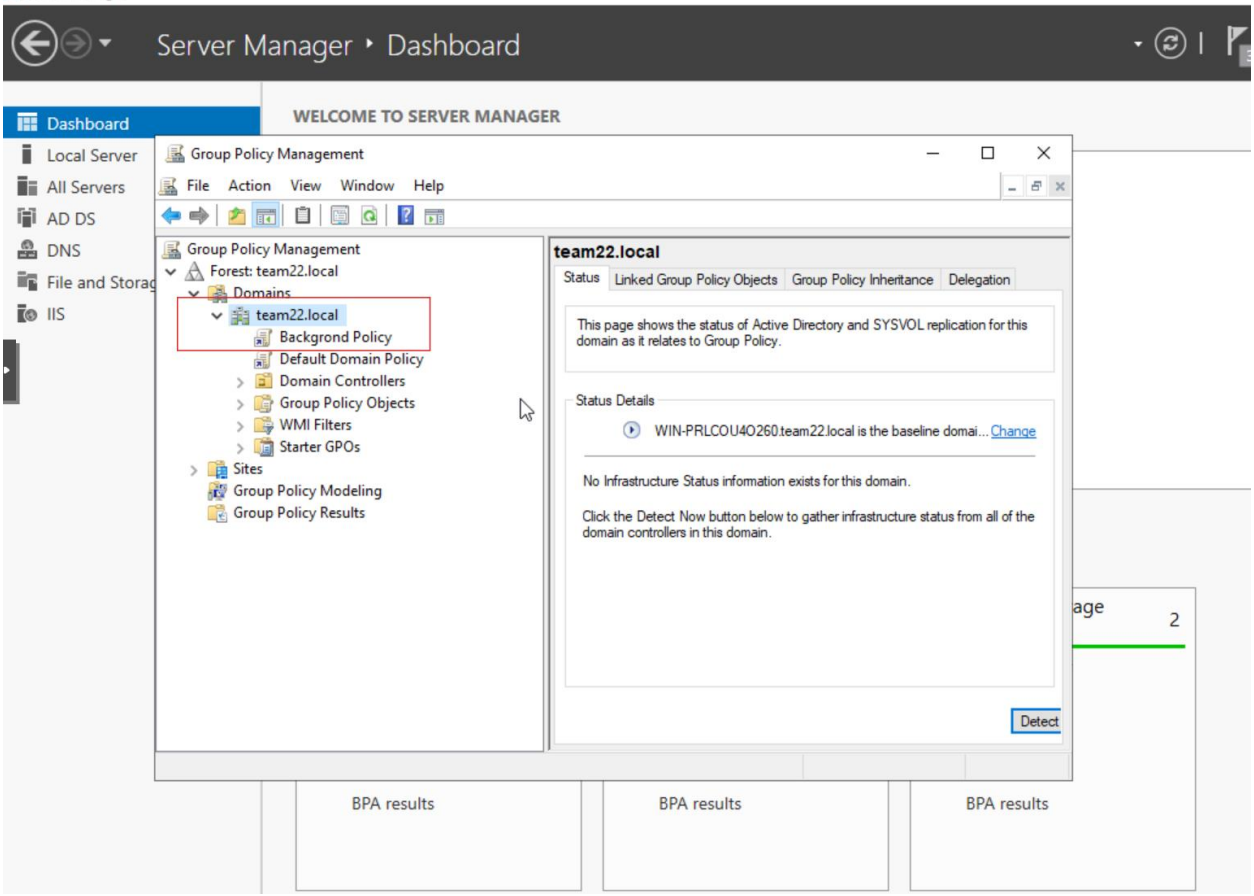

Figure 24: Displaying the Group Policy Management Console on the ADserver showing the newly created Background Policy linked under the team22.local domain. This confirms that the Group Policy Object was successfully created and applied within the domain hierarchy.

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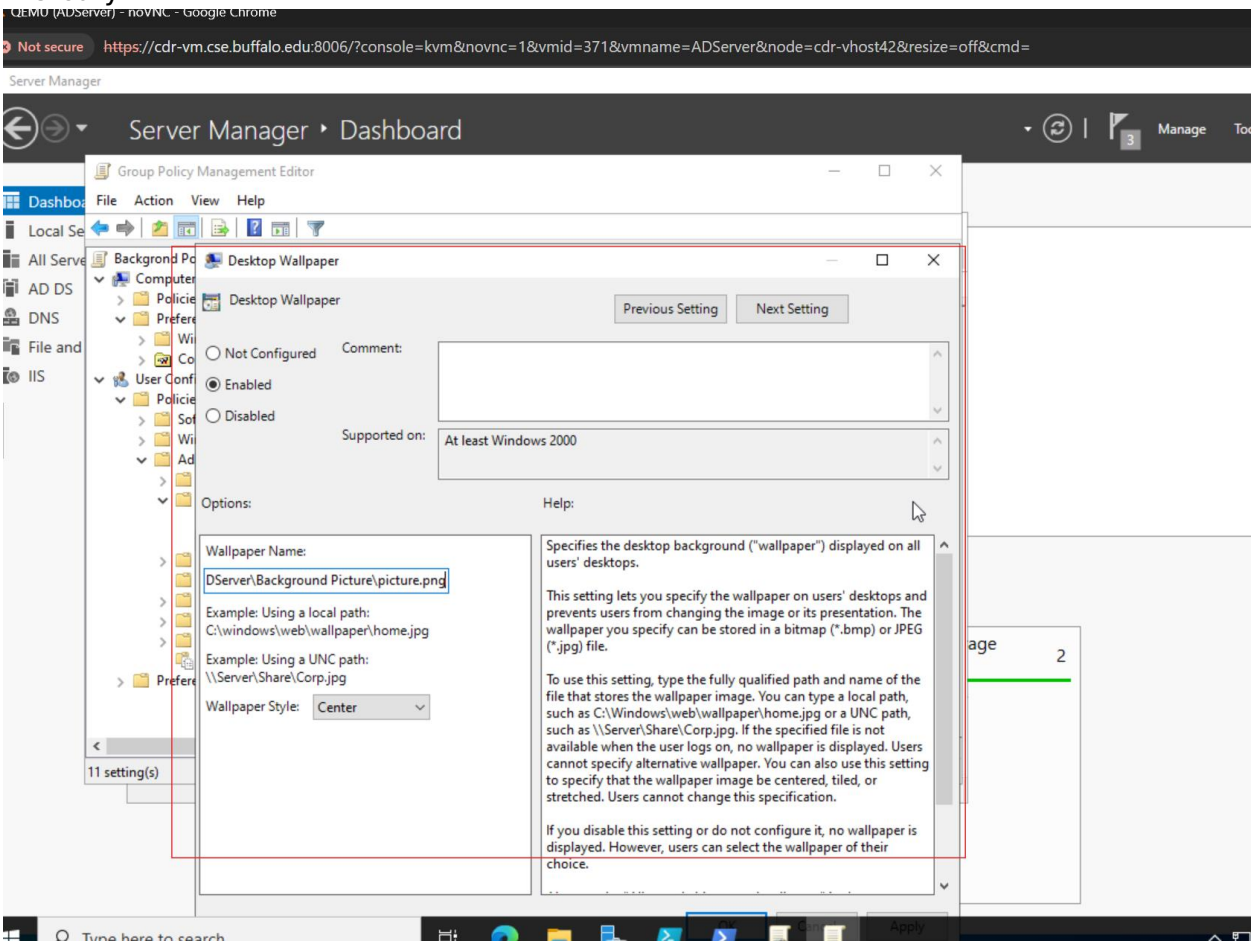

Figure 25: Editing the Background Policy Group Policy Object (GPO) to set the Desktio wallpaper configuration under User configuration - Administative template – desktop – desktop. The wallpaper path \ADSERVER\Background Picture\Pictue.png is applied to enforce a uniform desktop background across all domain users.

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Setup PowerShell Transcription Using a Group Policy

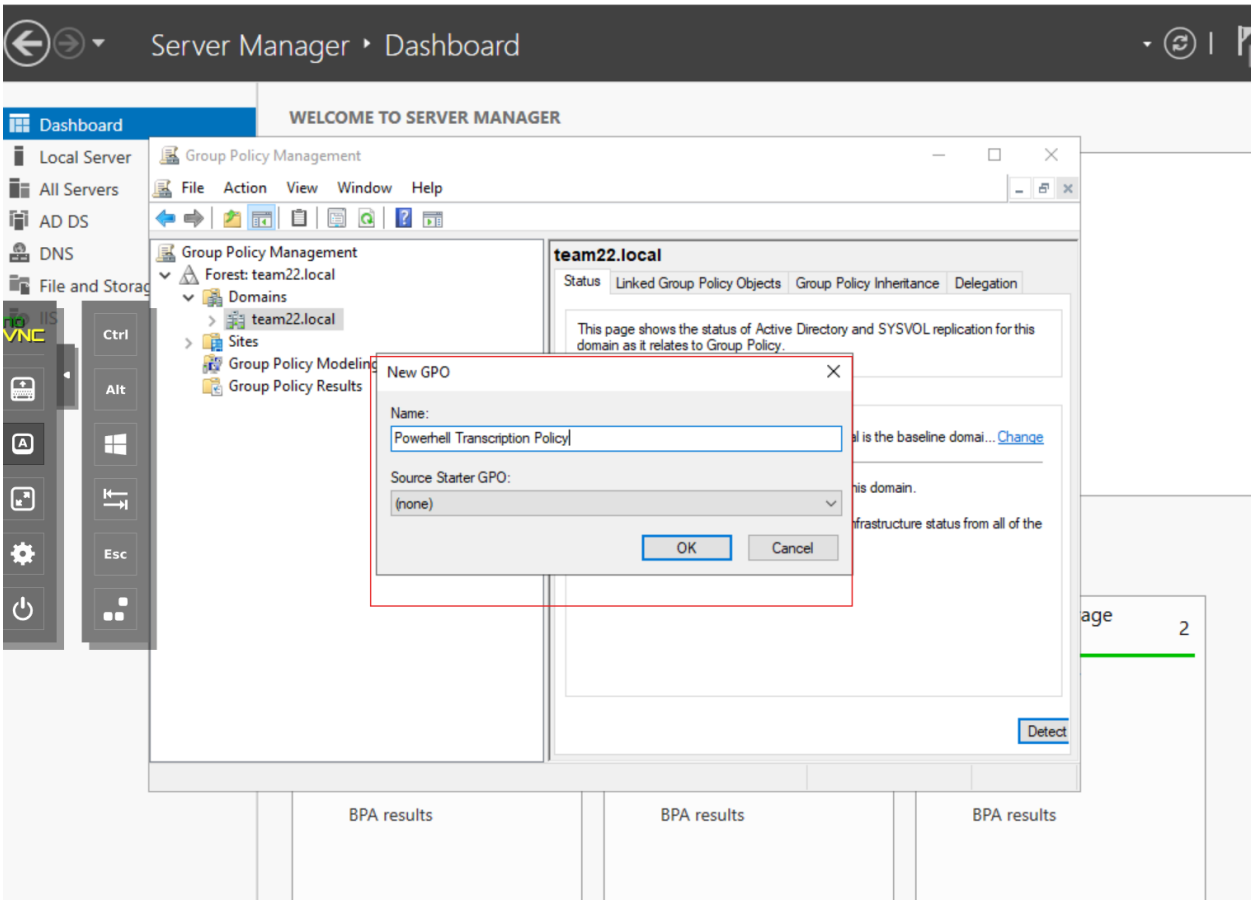

Figure 26: Creating a new Group Policy Object (GPO) named Powershell Transcription Policy within the team22.local domain using the Group Policy Management Console This policy will be used to enable and configure Powershell transcription for security auditing and activity logging purposes.

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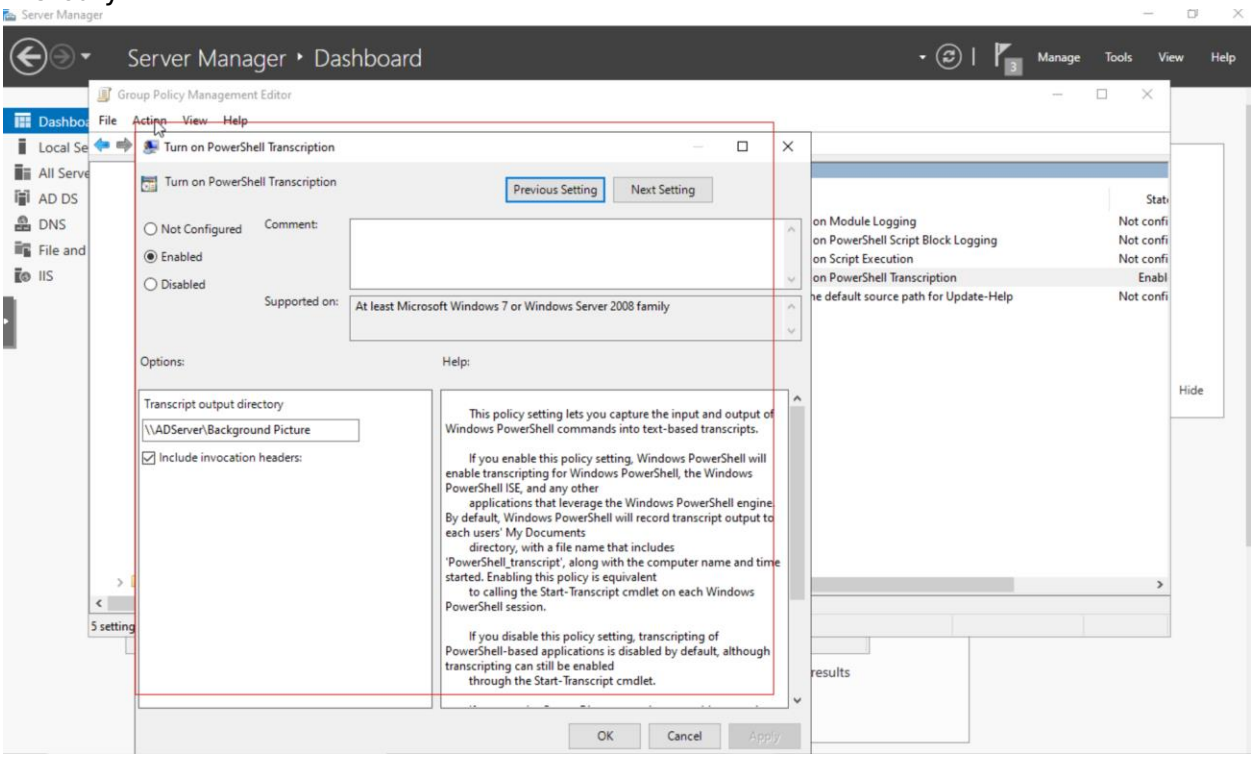

Figure 27:Enabling Powershell Transcription within the Powershell transcription policy gpo the transcription output directory is set to \ \Iserver|Background Picture, and the option to include invocation headers is checked this Ensures that all Powershell command activity ensures that all PowerShell command Activity across the domain system is logged for administrative review.

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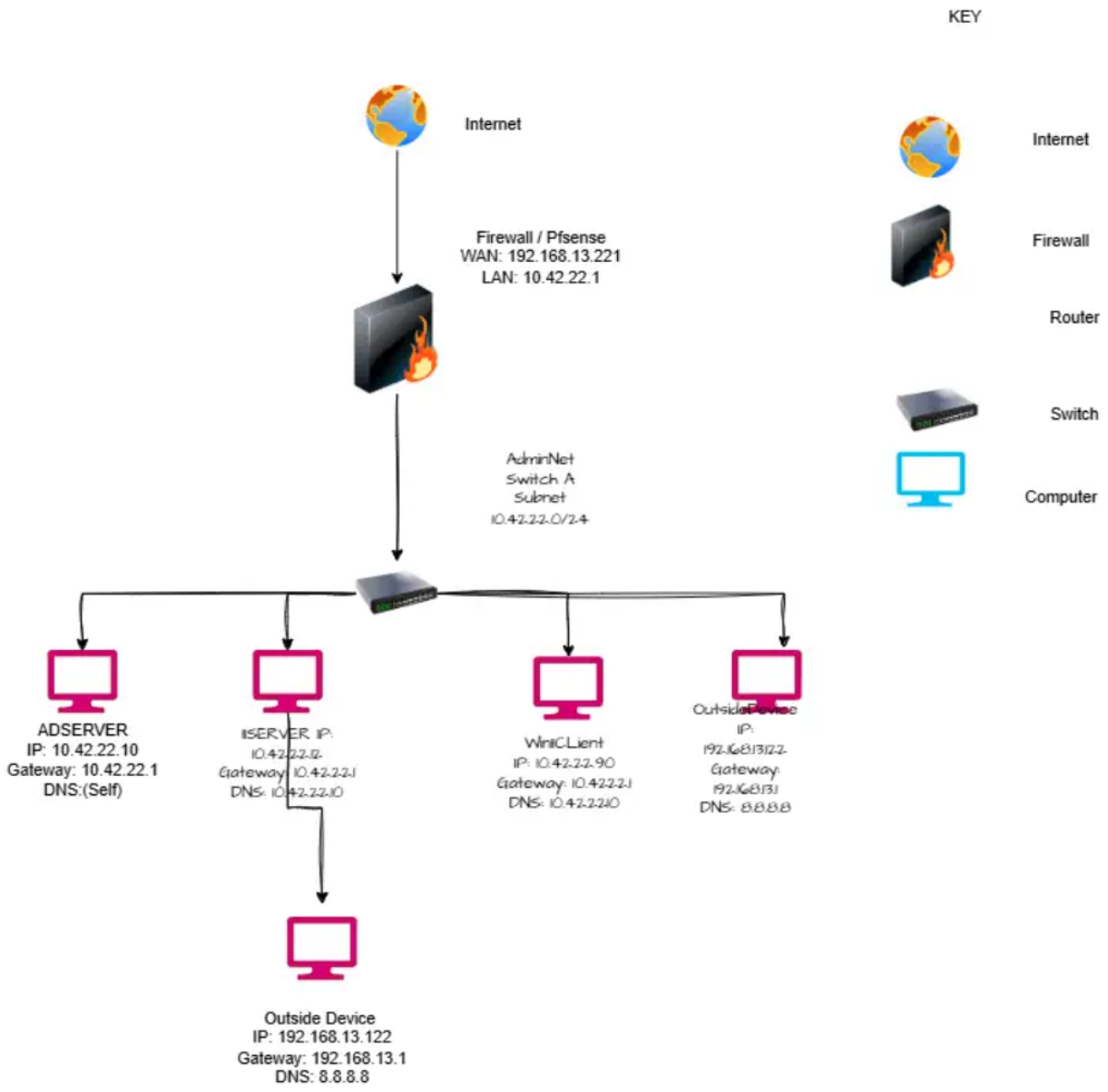
Topology